ENGINEERING OVERVIEW

MusicAID performance analysis MVP

A working reference-based system that turns a short audio or video performance into measure-level pitch and timing feedback.

3	30	30	< 60s
reference pieces	controlled fixtures	backend tests	MVP input limit

What was built

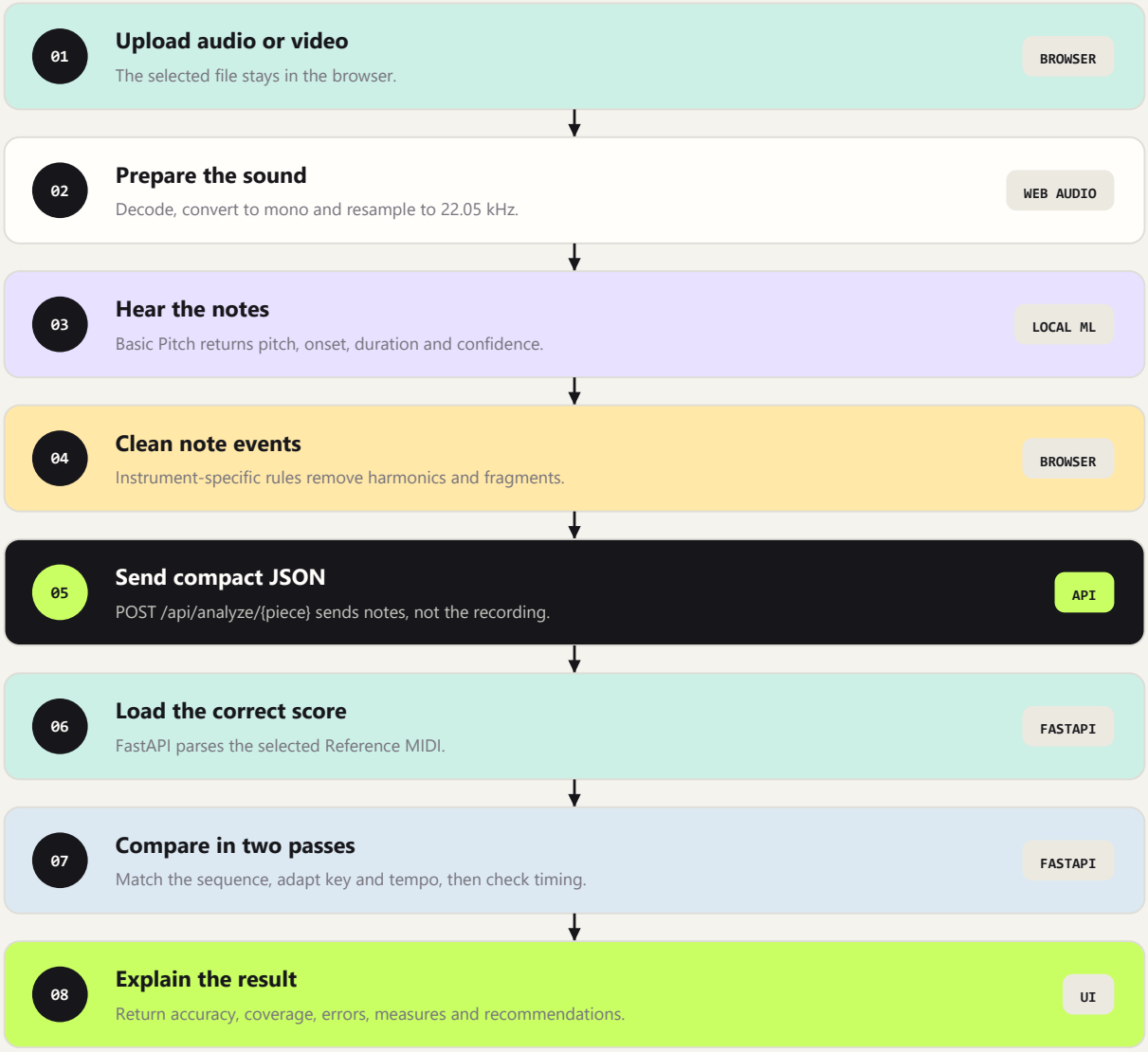
- Local audio-to-note transcription for audio and video uploads.
- Reference MIDI parsing with pitch, measure, beat and duration metadata.
- Tempo and transposition-aware sequence comparison.
- Wrong, missed, extra, timing and tuning-related feedback.
- A visual score, measure playback, guided error review and final recommendations.

> Transcription and evaluation are intentionally separated. The ML model determines what was heard; the analysis service determines how that performance relates to the selected score.

01 / FILE JOURNEY

One recording, from upload to feedback

The browser turns the recording into note events. FastAPI compares those events with the selected MIDI and returns an explainable report.



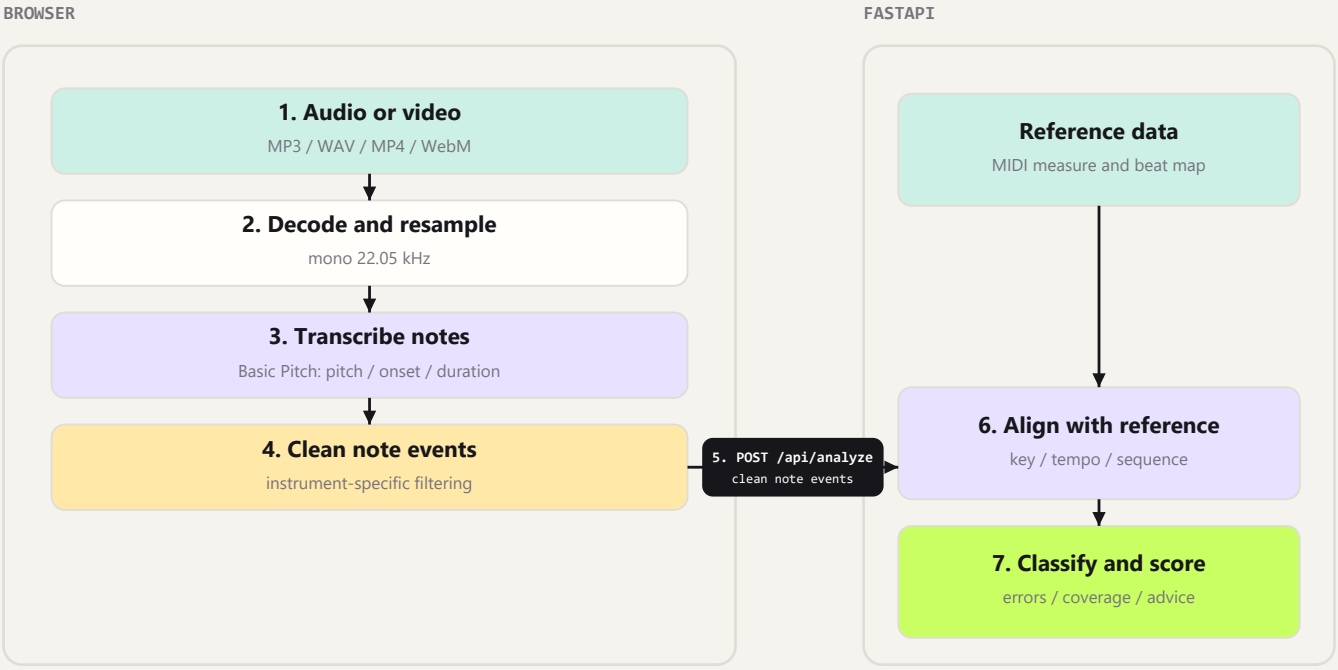
What actually crosses the API boundary

In the current MVP, the original recording stays in the browser. The backend receives only cleaned note events: pitch, start time, duration and confidence.

02 / SYSTEM ARCHITECTURE

A hybrid client and backend pipeline

The browser handles media and ML inference. FastAPI remains the source of truth for references, alignment, scoring and feedback.



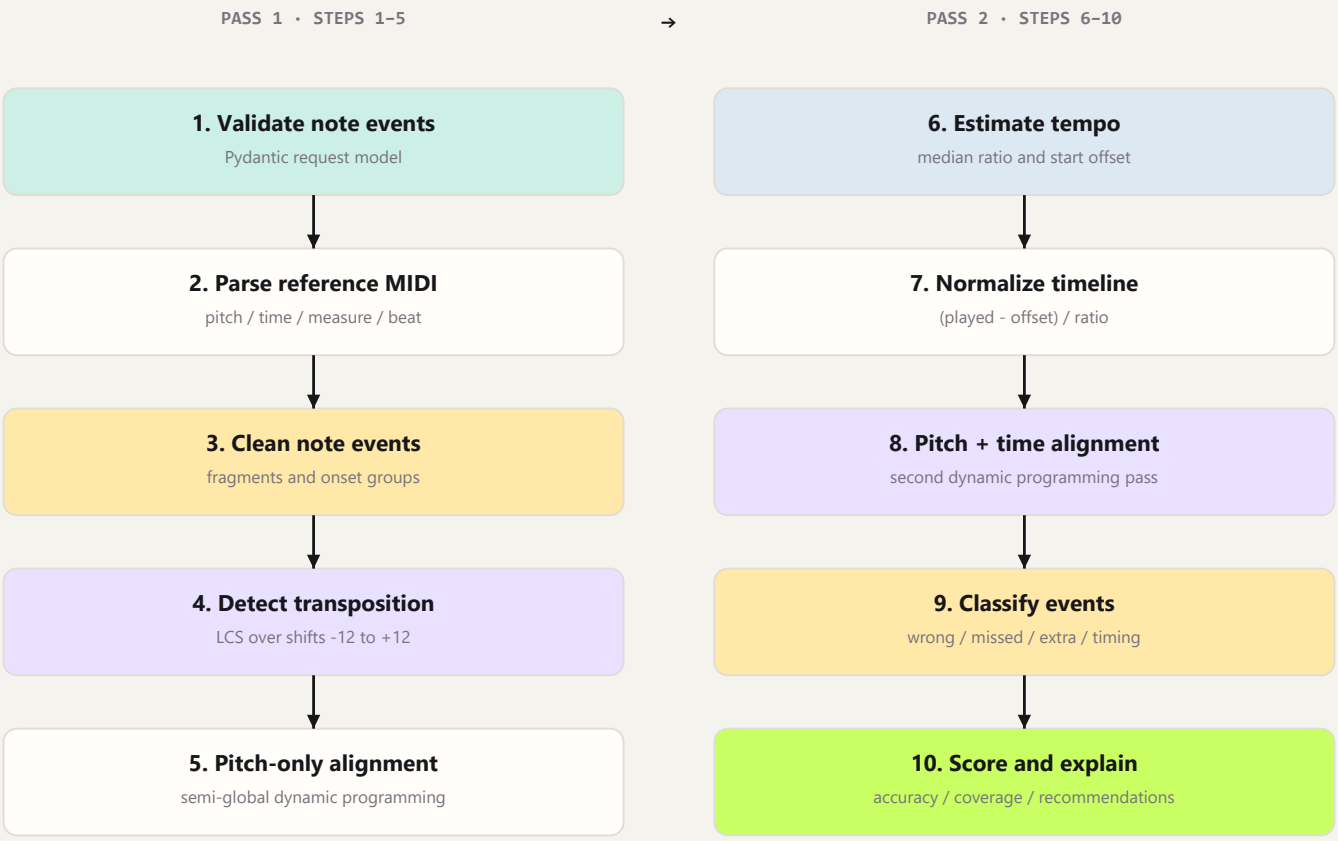
Design rationale

Local inference	No large media upload, lower server cost, immediate feedback and better privacy.
Server evaluation	One scoring implementation can serve web, mobile, teacher dashboards and future lesson rooms.
Small contract	The API receives only pitch, onset, duration and confidence. The transcription model can be replaced independently.

03 / BACKEND ALGORITHM

Two-pass alignment before error classification

The service first understands the musical sequence, then estimates tempo, normalizes time and performs a second comparison with timing context.



STEP 5 MATCHED NOTES → STEP 6 TEMPO ESTIMATION

RELIABILITY GATE

At least 3 exact pitch matches and 55% sequence agreement; otherwise return unrelated_piece.

Event	Interpretation	Penalty
Wrong / missed	A reference position has the wrong pitch or no onset.	1.0
Timing / extra	Correct pitch outside 160 ms, or an unmatched performed event.	0.5
Tuning	Repeated adjacent-semitone pattern on a recurring guitar pitch.	0.25

04 / TOOLS AND QUALITY CONTROLS

Simple components with explicit responsibilities

The stack was selected for a reproducible MVP, not for infrastructure volume.

Layer	Current choice	Why
Client	React, TypeScript, Vite	Fast UI iteration and typed analysis contracts.
Media	Web Audio API	Local decode, video audio extraction and resampling.
Transcription	Spotify Basic Pitch / TensorFlow.js	Open-source audio-to-note baseline that runs in the browser.
API	FastAPI and Pydantic	Explicit schemas, automatic docs and deterministic analysis.
Reference	MIDI and mido	Stable pitch, timing, measure and beat ground truth.
Validation	pytest plus controlled WAV fixtures	Separates transcription defects from comparison defects.

Instrument-specific transcription profiles

Profile	Keeps	Suppresses
Monophonic piano	One strongest note per onset	Overlaps, weak partials and short fragments
Polyphonic piano	Simultaneous chord tones	Weak resonances and duplicate sustained fragments
Guitar arpeggio	Sequential attacks with ringing strings	Harmonics, release transients and decay retriggers

100%
exact reference test

9 -> 6
guitar raw to real onsets

+/-12
semitone search range

160ms
timing tolerance

Controlled fixtures isolate two failure sources: transcription quality and comparison logic. Perfect, slower, transposed, partial, overplayed, wrong-note, late-note, mixed, very-poor and unrelated scenarios are generated for each reference.

05 / ENGINEERING DECISIONS

What stays, what changes and what is validated next

The MVP is intentionally narrow: it demonstrates the full analysis loop while keeping production risks visible.

Kept for production

A small note-event API, server-side scoring rules, reference MIDI as ground truth, instrument-specific preprocessing and controlled regression fixtures.

Changes for production

Add a reference registry and cache, a real evaluation dataset, local tempo warping for rubato, confidence calibration, observability, persisted attempts and optional server-side GPU transcription for weak devices or authoritative scoring.

Known boundaries

Basic Pitch returns quantized MIDI notes rather than cents. Dense pedal, reverb and complex polyphony remain harder than a clean monophonic phrase. The current MIDI parser assumes constant tempo references. Live WebRTC lessons and OMR are outside this test scope.

Production evolution

Current	Browser ML -> FastAPI alignment -> immediate feedback
Next	Evaluation corpus -> threshold calibration -> per-instrument metrics
Scale	Object storage -> job queue -> GPU workers -> persisted reports
Live	WebRTC audio -> streaming note events -> lesson state machine -> TTS feedback

The design does not hide model uncertainty. Transcription artifacts, musical mistakes, incomplete attempts and unsupported complexity remain distinguishable in both the code and the user feedback.